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United States Patent Application Publication

20250283343

Kind Code

A1

Publication Date

September 11, 2025

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SEISMIC WALL FUSE

Abstract

One or more seismic wall fuses that wrap around the edge of a building wall, the seismic wall fuses having a first wall panel on one side of the wall and a second wall panel on the other side of the wall, and an edge panel along the edge of the wall connected to the first and second wall panels. Each of the first and second wall panels have: a top portion that is attached to the wall, a mid portion that deforms to absorb seismic stresses, and a bottom portion that is also attached to the wall. In operation, the mid-portion of the first wall panel expands while the mid-portion of the second wall panel simultaneously contracts (or vice versa), thereby dissipating seismic energy in a direction perpendicular to the plane of the wall, which could otherwise topple the wall.

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Appl. No.: 18/598471

Filed: March 07, 2024

Publication Classification

Int. Cl.: E04H9/02 (20060101)

U.S. Cl.:

CPC E04H9/021 (20130101);

Background/Summary

TECHNICAL FIELD

[0001] The present invention relates in general to structures for dissipating seismic energy in a building, and in particular to structures for dissipating seismic energy in a concrete building wall.

BACKGROUND OF THE INVENTION

[0002] A variety of seismic devices exist for reinforcing different building structures and for helping to dissipate earthquake energy in buildings. Such devices may include various yielding devices and/or various vibration damping devices. A seismic fuse is an example of such a yielding device. In short, a seismic fuse is a steel assembly that is positioned between the ends of two beam sections and deforms as the beam sections move with respect to one another. Typically, a seismic fuse is installed with its two beam sections being connected to opposite diagonal corners of the frame of a wall. As such, the seismic fuse provides seismic reinforcement in directions in the plane of the wall itself.

[0003] Although seismic fuses are very useful, they are limited in that they only dissipate seismic energy across the plane of a wall. What would instead be desirable is to provide a seismic wall fuse assembly that would instead dissipate seismic loading in a plane perpendicular to the wall itself (for example, when one side of the wall is under tension and the other side of the wall is under compression). As such, the desired seismic fuse would dissipate energy in directions that would otherwise tend to topple over the wall.

[0004] In addition, it would also be desirable to provide such a seismic fuse system that is both fast and relatively easy to assemble. It would also be desirable to provide such a new seismic wall fuse system that can be easily retrofit into existing building walls, and especially onto concrete walls connected to building foundations. As will be shown, the present invention provides such a system.

SUMMARY OF THE INVENTION

[0005] The present invention provides a seismic wall fuse that is easy to attach onto the side edge of a building wall (such as a concrete building wall) to provide seismic energy dissipation. A unique advantage of the present system is that it resists seismic loading perpendicular to the plane of the wall (e.g.: when one side of the wall is under compression and the other side of the wall is in tension). As such, the present system resists seismic energy in the direction of the wall “toppling over”.

[0006] In preferred aspects, the present system provides a seismic wall fuse, comprising: a first wall panel and a second wall panel, wherein the first and second wall panels are positioned in parallel on opposite sides of an edge of the wall. Preferably, each of the first and second wall panels comprise: a top portion that is configured to be attached to the wall, a mid-portion that is configured to deform to absorb seismic stresses, and a bottom portion that is also configured to be attached to the wall. In operation, the mid-portion of the first wall panel deforms to expand while the mid-portion of the second wall panel simultaneously deforms to contract, and vice versa.

[0007] In various preferred aspects, the seismic wall fuse also includes an edge panel connected on one side to the first wall panel and on another side to the second wall panel. The edge panel is oriented perpendicular to the first and second wall panels such that the seismic wall fuse wraps around the edge of a wall.

[0008] In optional preferred embodiments, two or more of the present seismic wall fuses may be nested to wrap one around the other around the edge of the wall.

[0009] In further optional embodiments, a connection plate may be used to secure the seismic wall fuse to the building foundation. This connection plate may optionally be positioned in a nested configuration with one or more of the seismic wall fuses.

[0010] A unique advantage of the present system is that the same connectors that are to secure the first wall panel to one side of the wall may be used to secure the second wall panel to the opposite side of the wall. This may be achieved by having a plurality of wall connectors (such as bolts) passing through aligned holes in the first and second wall panels. In addition, the same connectors

can be used to secure the seismic wall fuse to a base plate connected to the building foundation (thereby securing the seismic wall fuse to the building foundation while avoiding the need for additional connectors and fasteners).

[0011] The mid-portions of the wall panels deform (i.e.: expand or contract) to dissipate seismic energy. In preferred aspects, the mid-portions of each wall panel comprises a plurality of laterally extending slits. These slits may preferably be diamond shaped. The mid-portions of these wall panels may also have a plurality of slots that are cut inwardly from opposite sides of the wall panel. These slots are preferably cut into opposite sides of the wall panel to widen when the wall panel is stretched, and to narrow when the wall panel is compressed.

[0012] In optional preferred aspects, the present system may also include a plurality of foundation connectors and base plate connecting the bottom portions of the wall panels down into a concrete wall foundation. Optionally, shear keys in the bottom of the wall may also be used.

[0013] In further optional aspects, the present system may be stacked such that one seismic fuse may positioned overtop of another one, running up along the edge of the wall.

[0014] Yet another advantage of the present system is that it provides a system that uses few materials, and limited parts. It is therefore fast and easy to fabricate and to install.

[0015] Moreover, in addition to being useful with concrete walls, the present system may also be used with wood, masonry, cross-laminated timber, etc. It is to be understood, therefore, that the present system is not limited to use solely with concrete walls and structures.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0016] FIG. 1 is an exploded perspective view of a pair of the present seismic wall fuses being nested together with a connection plate on a base plate, with the wall itself removed for clarity of illustration.

[0017] FIG. 2 is a view similar to FIG. 1, showing the connection plate and two of the present seismic wall fuses being nested together around the edge of a building wall, adding a portion of the wall itself.

[0018] FIG. 3A is a rear elevation view of a wall panel in a non-stressed position.

[0019] FIG. 3B is a front elevation view of the wall panel in a non-stressed position.

[0020] FIG. 4 is a front elevation view of the wall panel in an expanded state.

[0021] FIG. 5 is a front elevation view of the wall panel in a compressed state.

[0022] FIG. 6A is a top sectional plan view of a building wall showing the two of the present seismic wall fuses positioned at opposite ends of the building wall.

[0023] FIG. 6B is a close-up view similar to FIG. 6A, but showing only one of the present seismic wall fuses and its attachment to a connection and base plate.

[0024] FIG. 7A is a front elevation view of the present system installed on opposite sides of a building wall, showing the system's linkage to a building foundation, prior to any seismic loading of the wall or deformation of the present seismic fuse's wall panels.

[0025] FIG. 7B is similar to FIG. 7A, but shows movement of the wall in a shear direction (i.e.: parallel to the plane of the wall). FIG. 7B also shows corresponding deformation of each of the seismic wall fuses at opposite ends of the wall.

DETAILED DESCRIPTION OF THE FIGURES

[0026] The exploded view of FIG. 1 and the assembled view of FIG. 2 show a pair of seismic wall fuses **10A** and **10B** which are nested together, wrapping around the edge of a building wall. It is to be understood that the present system encompasses embodiments having only one seismic wall fuse **10**, a pair of nested seismic wall fuses **10A** and **10B**, or optionally three or more seismic wall fuses nested together, as desired.

[0027] As seen in the Figures, each wall fuse **10A** and **10B** preferably comprises a first wall panel **20** and a second wall panel **30**. As seen in FIGS. **2**, **6A** and **6B**, the first and second wall panels **20** and **30** are configured to be positioned in parallel on opposite sides of a wall **W**. As can also be seen in FIG. **2**, the seismic wall fuses **10A** and **10B** both wrap around the edge of wall **W** (in addition to being nested one inside the other, as shown).

[0028] In preferred aspects as seen in FIGS. **3A** and **3B**, first wall panel **20** comprises: a top portion **50** that is configured to be attached to the wall, a mid portion **52** that is configured to deform to absorb seismic stresses, and a bottom portion **54** that is configured to be attached to the wall.

[0029] Second wall panel **30** has a similar (e.g.: identical, but reversed/mirrored) shape to first wall panel **20**. In operation, for example, when seismic loading perpendicular to the plane of wall **W** acts in a direction to topple the wall over, the mid-portion **52** of the first wall panel **20** deforms to expand while the mid-portion **52** of the second wall panel **30** simultaneously deforms to contract. Conversely, the mid-portion **52** of the first wall panel **20** deforms to contract while the mid-portion **52** of the second wall panel **30** simultaneously deforms to contract. As a result, the seismic loading is resisted by the present seismic fuse on both sides of wall **W**.

[0030] FIG. **4** shows an example of the wall panel **20** pulled by seismic forces **F** to expand whereas FIG. **5** shows an example of the wall panel **20** with seismic forces **F** compressing the panel. (Note: FIGS. **4** and **5** are somewhat exaggerated views to clearly illustrate the deformation in the present system).

[0031] In preferred aspects as seen in FIGS. **1** and **2**, the first (i.e.: front) wall panel **20** and the second (i.e.: back) wall panel **30** are connected together by an edge panel **40** which is connected on one side to the first wall panel **20** and on another side to the second wall panel **30**. As can best be seen best in FIGS. **6A** and **6B**, edge panel **40** is oriented perpendicular to the first and second wall panels **20** and **30** such that the seismic wall fuse **10** can be positioned to wrap around the edge of wall **W** as seen in FIG. **2**.

[0032] In preferred aspects, a plurality of wall connectors **60** pass through the first and second wall panels **20** and **30** to connect the first and second wall panels to opposite sides of the wall. Preferably, each of the first and second wall panels **20** and **30** have apertures **51** passing therethrough, and the apertures **51** in each of the first and second wall panels **20** and **30** are preferably aligned such that the wall connectors **60** simply pass through the aligned apertures. Wall connectors **60** may be bolts or any other suitable form of connectors.

[0033] FIG. **1** also illustrates a base plate **100** and a connection plate assembly **110**. Base plate **100** is positioned under wall **W** (FIG. **2**), and seismic wall fuses **10A** and **10B** are nested inside connection plate assembly **110** (FIG. **2**). The same wall connectors **60** can pass through apertures in the connection plate assembly **110**, thereby using only a single bolt connector **60** to fasten seismic wall fuses **10** and **10B** and connector plate assembly **110** together. Connector plate assembly **110** is secured to base plate **100** which is in turn secured to foundation anchors **120**.

[0034] Further preferred details of the shape of wall panels **20** can be seen in FIGS. **3A** and **3B**. (Wall panel **30** is preferably a mirror image of wall panel **20**). The mid-portions **52** of each wall panel may preferably include a plurality of laterally extending slits **70**, which may optionally be diamond shaped. In addition, the mid-portions **52** of each wall panel **20** or **30** may comprise a plurality of slots **81**, **82** and **83** cut inwardly from opposite sides of the wall panel.

[0035] As can be seen in FIG. **4**, the wall panel deforms to absorb seismic stresses by widening slots **81**, **82** and **83** when the wall panel is stretched, or as can be seen In FIG. **5**, the wall panel deforms to absorb seismic stresses by narrowing the slots **81**, **82** and **83** when the wall panel is compressed.

[0036] As can also be seen, slots **81**, **82** and **83** can preferably be arranged such that: [0037] slot **81** separates the top portion **50** and the mid-portion **52** of the wall panel; [0038] slot **82** separates upper and lower regions of the mid-portion **52** of the wall panel; and [0039] slot **83** separates the mid-portion **52** and the bottom portion **54** of the wall panel.

[0040] As illustrated, slots **81** and **83** are cut inwardly from one side, whereas slot **82** is cut inwardly from the opposite direction.

[0041] It is to be understood that the illustration of slits **70** and slots **81** to **83** are merely exemplary and that the present invention is not limited to this specific illustrated embodiment. As such, other patterns of slots and slits are also encompassed within the scope of the present invention.

[0042] FIGS. **6A** and **6B** illustrate top sectional plan views of a pair of the present seismic wall fuses wrapping around the edges of a wall **W**. The passage of connector bolts **60** through wall **W** and through wall panels **20** and **30** and through connector plate assembly **110** can be seen.

[0043] FIG. **6A** further illustrates an optional shear key **150** in wall **W** and base plate **100** may have a member (which may be steel) **160** that is received up into the bottom of shear key **150**. This shear key structure is also seen in FIGS. **7A** and **7B** described below.

[0044] FIGS. **7A** and **7B** specifically illustrate the case of seismic loading in a shear direction in the plane of wall **W**. Note, this direction of loading is perpendicular to the “toppling” loading deformations described in the other Figures. As seen in FIG. **7A**, foundation anchors **120** are received down into foundation **130**. Foundation **130** may optionally be a concrete foundation and foundation anchors **120** may optionally be steel rebar. FIG. **7A** illustrates the case of no seismic loading. FIG. **7B** shows the case of seismic shear loading. As can be seen, a seismic wall fuse **10A** (which may optionally comprise a nested pair of seismic wall fuses **10A** and **10B** similar to FIGS. **1** and **2**) will expand (to the state/position shown in FIG. **5**) whereas an opposite a seismic wall fuse **10B** (which may also optionally comprise a nested pair of seismic wall fuses **10A** and **10B** similar to FIGS. **1** and **2**) will be compressed (to the state/position shown in FIG. **4**). In this particular case, wall panels **20** and **30** in seismic wall fuse **10A** will be expanded together and wall panels **20** and **30** in seismic wall fuse **10B** will be compressed together. As can be appreciated, therefore the present system can therefore resist seismic loading in both perpendicular directions (i.e.: the direction perpendicular to the wall that would otherwise topple the wall and in the shear direction in the plane of the wall). This feature of the present invention is very advantageous in that seismic loading is often in a direction that is at an oblique angle to the wall (i.e.: a direction that has components both parallel and perpendicular to the plane of the wall). Simply put, the present system can resist seismic loading in all directions.

[0045] Lastly, it is to be understood that the presently described and claimed invention is not limited solely to the embodiments described herein but also covers embodiments and variations within the scope of knowledge of a person skilled in the art.

Claims

1. A seismic wall fuse, comprising: a first wall panel, and a second wall panel, wherein the first and second wall panels are configured to be positioned in parallel on opposite sides of a wall, and wherein each of the first and second wall panels comprise: a top portion that is configured to be attached to the wall, a mid portion that is configured to deform to absorb seismic stresses, and a bottom portion that is configured to be attached to the wall.
2. The seismic wall fuse of claim 1, wherein: the mid-portion of the first wall panel deforms to expand while the mid-portion of the second wall panel simultaneously deforms to contract, and the mid-portion of the first wall panel deforms to contract while the mid-portion of the second wall panel simultaneously deforms to contract.
3. The seismic wall fuse of claim 1, further comprising: an edge panel connected on one side to the first wall panel and on another side to the second wall panel, wherein the edge panel is oriented perpendicular to the first and second wall panels such that the seismic wall fuse can be positioned to wrap around the edge of a wall.
4. The seismic wall fuse of claim 1, further comprising: a plurality of wall connectors passing through the first and second wall panels to connect the first and second wall panels to opposite

sides of a wall.

5. The seismic wall fuse of claim 4, wherein the first and second wall panels have apertures passing therethrough and wherein the apertures in the first and second wall panels are aligned such that the wall connectors pass through the aligned apertures in both the first and second wall panels.
 6. The seismic wall fuse of claim 5, wherein the wall connectors are bolts.
 7. The seismic wall fuse of claim 1, wherein the mid-portion of each wall panel comprises: a plurality of laterally extending slits.
 8. The seismic wall fuse of claim 7, wherein the laterally extending slits are diamond shaped.
 9. The seismic wall fuse of claim 1, wherein the mid-portion of each wall panel comprises: a plurality of slots cut inwardly from opposite sides of the wall panel.
 10. The seismic wall fuse of claim 9, wherein the wall panel deforms to absorb seismic stresses by: widening the slots when the wall panel is stretched, and narrowing the slots when the wall panel is compressed.
 11. The seismic wall fuse of claim 9, wherein the slots comprise: a first slot separating the top portion and the mid-portion of the wall panel; a second slot separating upper and lower regions of the mid-portion of the wall panel; and a third slot separating the mid-portion and the bottom portion of the wall panel.
 12. The seismic wall fuse of claim 11, wherein the first and third slots are cut inwardly in a first direction and the second slot is cut inwardly in a second direction, the first and second directions being opposite to one another.
 13. The seismic wall fuse of claim 1, further comprising: a wall, wherein the first and second wall panels are connected to opposite sides of the wall.
 14. The seismic wall fuse of claim 13, wherein the wall is made of concrete.
 15. The seismic wall fuse of claim 13, further comprising: a base plate connected to the bottom of the wall; and a connection plate assembly connected both to the base plate and to the first and second wall panels.
 16. The seismic wall fuse of claim 15, further comprising: a plurality of foundation anchors connecting the base plate to a building foundation.
 17. The seismic wall fuse of claim 15, further comprising: a shear key connected to the base plate, wherein the shear key is receivable into an aperture in the wall.
 18. The seismic wall fuse of claim 1, wherein a pair of the seismic wall fuses are nested together around the edge of a building wall.
 19. The seismic wall fuse of claim 18, further comprising: a connection plate assembly nested around the pair of seismic wall fuses.
 20. The seismic wall fuse of claim 13, wherein the wall is made of: wood, masonry, or cross-laminated timber.
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